

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EE)

May 2014

EE203/EE203.01 Electrical Measurement & Measuring Instruments

Date: 07.05.2014, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain briefly working of single phase Electrodynamometer power factor meter. [4]
(b) Explain the need of synchroscope and phase sequence indicator. [3]
- Q - 2 (a) Draw Equivalent circuit and phasor diagram of Potential Transformer. Derive the equation of phase angle for Potential Transformer. [8]
(b) Explain in detail Varley loop test for localization of cable faults. [6]

OR

- Q - 2 (a) Draw the phasor diagram of Current Transformer and derive the equation for transformation ratio of Current Transformer. [7]
(b) A potential transformer, ratio 1000/100 volt, has the following constants: [7]
Primary resistance = 94.5Ω
Secondary resistance = 0.86Ω
Primary reactance = 66.2Ω
Total equivalent reactance = 110Ω
No load current = 0.02 A at 0.4 power factor.
Calculate: (1) Phase angle error at no load (2) burden in VA at unity power factor at which the phase angle will be zero.
- Q - 3 (a) Why scale of Moving Iron instrument is non-uniform? Derive the general torque equation of Moving Iron instrument. [7]
(b) A moving coil voltmeter with a resistance of 20Ω gives a full scale deflection of 120° [7]
when a potential difference of 100 mV is applied across it. The moving coil has dimensions of $30 \text{ mm} \times 25 \text{ mm}$ and is wound with 100 turns. The control spring constant is $0.375 \times 10^{-6} \text{ Nm/deg}$. Find the flux density in the air gap. Find also the diameter of the copper wire of coil winding if 30 percent of instrument resistance is due to coil winding. The specific resistance for copper = $1.7 \times 10^{-8} \Omega\text{m}$.

OR

- Q - 3 (a) Derive the force and torque equations of Electrostatic instruments. State advantages and disadvantages of Electrostatic instruments. [7]
- (b) In an electrodynamic instrument the total resistance of the voltage coil circuit is $8.2 \text{ K}\Omega$ and the mutual inductance changes uniformly from $-173 \mu\text{H}$ at zero deflection to $+175 \mu\text{H}$ at full scale, the angle of full scale being 95° . If a potential difference of 100 V is applied across the voltage circuit, and a current of 3 A at a power factor of 0.75 is passed through the current coil, what will be the deflection, if the spring control constant is $4.63 \times 10^{-6} \text{ N-m/rad}$? [7]

SECTION - II

- Q - 4 (a) Explain the following terms related to measuring instruments: [4]
- Reproducibility
 - Sensitivity
- (b) Explain briefly relative limiting error [3]
- Q - 5 (a) Explain any two methods for measurement of medium order resistance. [7]
- (b) Draw the phasor diagram and derive balance equation of Schering bridge. [7]

OR

- Q - 5 (a) Explain direct deflection and loss of charge method for measurement of high order resistance. [7]
- (b) A bridge consists of the following: [7]
- Arm ab: a choke coil having a resistance R_1 and inductance L_1 .
- Arm bc: resistance R_3
- Arm cd: capacitor C_4 in series with a resistance R_4
- Arm da: resistance R_2
- When this bridge is fed from a source of 500 Hz , balance is obtained under following conditions:
- $R_2 = 2410 \Omega$; $R_3 = 750 \Omega$; $C_4 = 0.35 \mu\text{F}$; $R_4 = 64.5 \Omega$. The series resistance of capacitor $= 0.4 \Omega$. Derive the balance equation for this bridge. Calculate the resistance and inductance of the choke coil. The supply is connected between a and c and the detector between b and d.

- Q - 6 (a) Explain the working of slide wire D.C. potentiometer. Also, explain standardization of potentiometer. [7]
- (b) Explain different methods for determination of B-H curve of a specimen. [7]

OR

- Q - 6 (a) Explain in detail the working of Co-ordinate type A.C. potentiometer. [7]
- (b) Explain in detail magnetic potentiometer. [7]